



Phi

A constructed language

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1 Introduction

Phi is a constructed language. A constructed language is one whose phonology, grammar, and vocabulary, instead of having developed naturally, are consciously devised¹. The sound of the name *Phi* embodies the simplicity and softness of the language itself. *Phi* (Φ) is also beauty in mathematics. It is

“the golden ratio, an irrational number with a value of approximately 1.618033988 which expresses the relationship that the sum of two quantities is to the larger quantity as the larger is to the smaller.” – Wiktionary²

It is thought that the value of *phi* is one of the fundamental characteristics of nature; in the language Phi, nature is fundamental to its being. Listening to spoken Phi one is sometimes reminded of the sounds of the natural world. There is also a musicality and rhythm to sentences in the language.

1 https://en.wikipedia.org/wiki/Constructed_language

2 <https://en.wiktionary.org/wiki/%CF%86>

2 The Tengwar

Phi can be written in either of two ways: using the standard Latin alphabet or the Tengwar script. Tengwar was invented by J. R. R. Tolkien and described first in Appendix E of the third book in his trilogy, *The Lord of the Rings*. There are numerous sources of information available concerning the Tengwar so details of the script will not be explored here.

The look and feel of the Tengwar characters (*tengwa*) and its diacritical marks (*tehta*) match the ideals and philosophies behind this language. Phi has a small set of consonants and vowels so it is relatively straightforward to map its graphemes to Tengwar characters. The Unicode font *Tengwar Telcontar*³ is used in this manual to render Phi text in Tengwar script. A custom keyboard layout⁴ was employed to facilitate typing the Tengwar characters.

Table 1 outlines the mapping between the graphemes used in Phi and corresponding Tengwar characters, as well as the keys necessary⁵ to produce the characters using the font and keyboard layout mentioned previously. An application or utility that supports Graphite fonts⁶ is required for the font to render correctly. This manual was produced using LibreOffice Writer v7.1.2.2, which renders *Tengwar Telcontar* properly. A link to details of this requirement can be found in the first footnote below.

Table 1: Phi grapheme to Tengwar mapping

Grapheme	Tengwar	Keypress
p	ᑭ	p
t	ᑮ	t
m	ᑭᑭ	m

3 <http://freetengwar.sourceforge.net/tengtelc.html>

4 <http://freetengwar.sourceforge.net/keylayouts.html>

5 Keypresses for the diacriticals are given for macOS. Windows uses ctrl-alt instead of option.

6 https://scripts.sil.org/cms/scripts/page.php?site_id=projects&item_id=graphite_fonts

n	ᄎ	n
r	ᄏ	shift-r
s	ᄑ	s
s	ᄒ	shift-s
h	ᄔ	h
l	ᄖ	l
w	ᄗ	w
ph	ᄙ	shift-p
th	ᄛ	shift-t
sh	ᄝ	shift-c
wh	ᄞ	shift-k
i	ᄠᄡ	i / option-i
e	ᄢᄣ	e / option-e
o	ᄤᄥ	o / option-o
a	ᄧᄨ	a / option-a

3 Examples

Following are several examples of Phi phrases. They are given first in English, then in romanized text, and finally in Tengwar script.

My name is Daniel

phiathe soa mia na thanie te phi

ḥḥ.ſ.ṁ.ḥṁ.ṛ.ḥ

I love you.

mia na thi te lothea

ṁ.ṁ.ḥ.ṛ.ḥ

The sky is blue.

whetho na mipho te phi

Ḑḥ.ṁ.ṁ.ḥ.ṛ.ḥ

They like food.

lo sha na noshea te lephe

ḥ.ḥ.ṁ.ṁ.ḥ.ṛ.ḥ

I like it.

mia na sha te lephe

ṁ.ṁ.ḥ.ṛ.ḥ

I do not like green!

ho mia na hashe te me lephe

Ḑ.ṁ.ṁ.Ḑ.ṛ.ṁ.ḥ

mia na miwho soa mia te li lephe

noshea na soa lo mia te phi

noshea soa lo mia na shishoa te phi

wha thi na ropo shishoa te la phe

lo mia na lo lipha sathe te lothea

ፎ.ጠ.ጠ.ፎ.ፎ.ፎ.ፎ.ፎ.ፎ.ፎ.ፎ.